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Python Programming (20 Hours)

Section 1: Python Basics (2 hrs)

- Introduction to Python programming
- Installation and Environment Variables setting
- Statements and Syntax
- Variables Assignment
- Identifiers
- Basic Inputs
- Numbers
- Operators

Section 2: Conditionals and Loops (3 hrs)

- If else statement
- If elif else statement
- Switch
- While loop
- For loop
- Break, Continue, Pass

Section 3: Sequences: String, List and Tuple , Mapping and Set types (3 hrs)

- Sequences
- Strings and operators
- String built-in Functions
- List and methods
- Tuples
- Mapping Type: Dictionaries
- Mapping type built-in functions
- Set Type –operators, built-in methods

Section 4: Functions and Modules (3 hrs)

- What are Functions?
- Calling Functions
- Creating Functions
- Formal Arguments
- Variable Scope
- Recursion
- Modules introduction
- Modules and files
- Importing Modules
- Module Built-in Functions
- Packages

Section 5: Object-Oriented Programming (3 hrs)

- Object-Oriented Programming
- Classes
- Class Attributes
- Instances
- Instances and Attributes
- Binding and Method Invocation
- Static Methods and Class Methods
- Inheritance

Section 6: File handling (read/write CSV, JSON) (2 hrs)

- Basic File Handling Concepts
- File Built-in Functions
- Handling CSV Files
- Handling JSON Files

Section 7: NumPy (2 hrs)

- Introduction to NumPy
- NumPy arrays and operations
- Random Number Functions in NumPy
- Introduction to pandas
- DataFrames and Series in Pandas
- Reading and Writing data in Pandas
- Data manipulation: filtering, grouping, merging

Section 8: Data Visualization (2 hrs)

- Introduction to Matplotlib and Seaborn
 - Line plots, bar charts, histograms, scatter plots
 - Customizing plots
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Data Science (25 Hours)

Section 1: Introduction to Data Science (3 hrs)

- What is Data Science?
- Data Science lifecycle
- Data Analyst vs Data Scientist
- Real-world applications

Section 2: Data Wrangling and Preprocessing (6 hrs)

➤ Handling Missing Values

- Techniques: drop, fill, imputation (mean, median, mode)
- Using pandas methods (`isnull()`, `fillna()`, `dropna()`)

➤ Removing Duplicates

- Identifying and removing duplicate rows (`drop_duplicates()`)

➤ Encoding Categorical Variables

- One-hot encoding
- Label encoding
- When and where to use each

➤ Feature Scaling

- Standardization (Z-score)
- Normalization (Min-Max scaling)
- When scaling is necessary (e.g., KNN, SVM)

➤ Outlier Detection & Treatment

- Using IQR and Z-score methods
- Box plot interpretation
- Handling outliers (removal, capping)

Section 3: Exploratory Data Analysis (EDA) (6 hrs)

➤ Univariate Analysis

- Frequency tables
- Bar plots, histograms
- Box plots for distribution and outliers

➤ Bivariate Analysis

- Scatter plots
- Correlation matrix
- Grouped box plots

➤ Multivariate Visualization

- Heatmaps
- Pair plots using Seaborn

➤ Correlation and Covariance

- Pearson vs Spearman correlation
- Visualizing correlation with heatmaps

➤ Using Tools

- Seaborn, Matplotlib, Pandas Profiling for EDA
- Hands-on profiling report generation

Section 4: Statistical Foundations (6 hrs)

➤ Descriptive Statistics

- Mean, median, mode
- Variance, standard deviation, range, IQR

➤ Probability Basics

- Basic probability rules
- Probability distributions: normal, binomial, uniform
- Random variables

➤ Inferential Statistics

- Sampling and sampling distribution
- Hypothesis testing (Z-test, T-test)
- P-values and significance levels
- Confidence intervals: concept and interpretation

Section 5: Feature Engineering & Selection (4 Hours)

➤ **Feature Engineering**

- Creating new features from existing ones
- Extracting date, time, and text-based features
- Binning and transformations (log, square root)

➤ **Feature Selection**

- Filter methods (correlation, variance threshold)
- Wrapper methods (recursive feature elimination)
- Embedded methods (Lasso regularization)

➤ **Dimensionality Reduction (Intro Only)**

Machine Learning Basics (20 hrs)

Section 1: Introduction to Machine Learning (1.5 Hours)

- What is Machine Learning?
- Types of Machine Learning
 - Supervised vs Unsupervised Learning
 - Brief intro to Reinforcement Learning
- Real-world ML use cases

Section 2: Scikit-learn & ML Workflow Basics (1.5 Hours)

- Overview of Scikit-learn
- Steps in a typical ML workflow:
 - Importing data
 - Preprocessing
 - Model training and evaluation
- Hands-on: Basic classification/regression pipeline using Scikit-learn

Section 3: Supervised Learning Algorithms (9 Hours)

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machines (SVM)

Section 4: Unsupervised Learning (5 Hours)

- K-Means Clustering
- Hierarchical Clustering
- PCA (Dimensionality Reduction)

Section 5: Model Evaluation & Validation (3 Hours)

- Train/test split
 - Cross-validation
 - Classification Metrics
 - Accuracy, Precision, Recall, F1-score, ROC-AUC
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Artificial Intelligence (25 Hours)

Section 1: Introduction to AI (1 hrs)

- What is AI?
- History and evolution of AI
- AI vs ML vs DL
- AI subfields (NLP, CV, Robotics, etc.)

Section 2: Search Algorithms & Problem Solving (6 hrs)

- State space search
- BFS, DFS
- Heuristics and A* Algorithm
- Constraint satisfaction problems

Section 3: Knowledge Representation & Reasoning (3 hrs)

- Logic-based representation
- Propositional and First-order logic
- Inference rules, truth tables

Section 4: Natural Language Processing (6 hrs)

- Text preprocessing: tokenization, stemming, lemmatization
- Bag of Words, TF-IDF
- Sentiment analysis
- Named Entity Recognition (NER)

Section 5: Neural Networks & Deep Learning Basics (5 hrs)

- Perceptron and activation functions
- Feedforward networks and backpropagation
- Keras/TensorFlow intro
- Build and evaluate basic neural networks

Section 6: AI Applications and Trends (4 hrs)

- AI in healthcare, finance, robotics, etc.
- Ethical considerations in AI
- Tools: ChatGPT, Google AI
- Career trends and opportunities

Common Project (15 Hours)